Factory Reset: How many Branches are too Many?

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ABSTRACT

This research investigated the best number of unique branches or pathways through a branching narrative game that we are designing in order to ensure players feel immersed and engaged. Branching narrative games have many different choices the player can make which lead to different paths through the graph that is the entire story. The number of these unique paths can vary depending on which story threads designers feel are important. To explore what amounts of these pathways might be best, we had two treatment groups: one playing through a version of our demo with a high number of unique branches, and the other playing through a version of our demo with a low number of unique branches. Both groups were instructed to play through for the same amount of time and were then given an exit interview along with a verified instrument survey of Player Experience Inventory (PXI). Our findings indicate that the participants in the high-branching group felt more satisfied with the number of branches in our game than the participants in the low-branching group.

Keywords

game design, player agency, moral decision-making, design-based research, ethical gameplay, narrative design, branching narrative, interactive narrative

INTRODUCTION

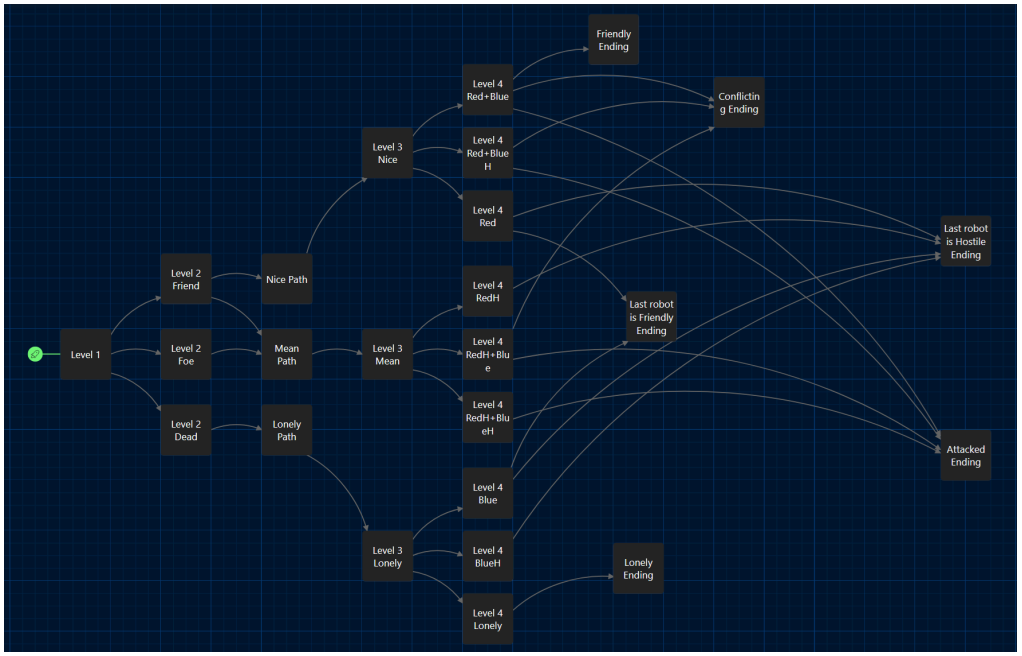
Branching narrative is a popular genre in video games. Games that employ this have a narrative that reacts and changes according to what the player chooses to do by having the narrative split at a decision point, creating a branching tree-like structure. Games like Baldur's Gate 3 (Larian Studios 2023) and Detroit: Become Human (Quantic Dream 2018) both employ this technique. Detroit: Become Human employed a very classical approach to this technique, even going so far as to have the narrative and decision trees available to the player to look at after completing each level. Baldur's Gate 3 employed an extended 6 year long development period in

order to bring a very wide spanning narrative to the players with thousands of different paths through the tree. It also is structured slightly differently to other branching narrative games: since the player can approach gameplay from any angle in the semi-open world, the individual sections of narrative within an act are designed in such a way that they can be interwoven with each other, creating almost a web with the destination of the plot at the center as opposed to a tree.

Regardless of the specific implementation of branching narrative in a game, there are always a few factors that are important to keep in mind: the number of paths through the narrative tree, the exponential development and design time, and the narrative importance of each decision. We aim to find a balance between these factors in the specific context of the game we are designing, Factory Reset. Our hypothesis is that a balance point of the number of unique branches through a narrative is a moderate amount, in between the low branching and high branching versions of our demo. We aim to find this by changing the number of pathways in each treatment group and measuring their feelings of immersion and agency. By keeping our number of branches at this hypothetical balance point, this assists in ensuring each decision point or branch is significant enough to feel important to the player, as we have more focus and attention to spend on each branch.

Branching Narrative In Our Game

The main point of measurement for this research is the number of unique pathways through the narrative decision tree of our game. This differs from the number of endings, as different players may reach the same ending while making slightly different choices throughout the game. In the design of our narrative for this game, we focused on hand crafting each branch to fit with the narrative specifically. Some decision points lead to more than two branches, while some others lead to a single path. This depends on the specific context of what is occurring in the game at that point as well as what the actual decision is that is being made by the player.



The narrative tree for our high-branching version.

GAME PROTOTYPE DESCRIPTION

Factory Reset is a narrative-driven puzzle game that explores moral choice and systemic consequence through its interactive design. The game is set in a malfunctioning robot factory, where the player assumes the role of a human inspector tasked with diagnosing and managing defective robots. What begins as a series of technical evaluations gradually unfolds into a moral investigation as the robots exhibit signs of autonomy, and self-preservation. The prototype employs a 2D top-down grid layout, emphasizing spatial reasoning and deliberate movement. Players navigate the factory environment using WASD controls to explore and reposition within each puzzle space, while the Space key enables interaction with nearby NPC robots. During these interactions, the player can choose to Talk or Hack. The choices made between these two interaction modes influence the direction of the branching narrative.

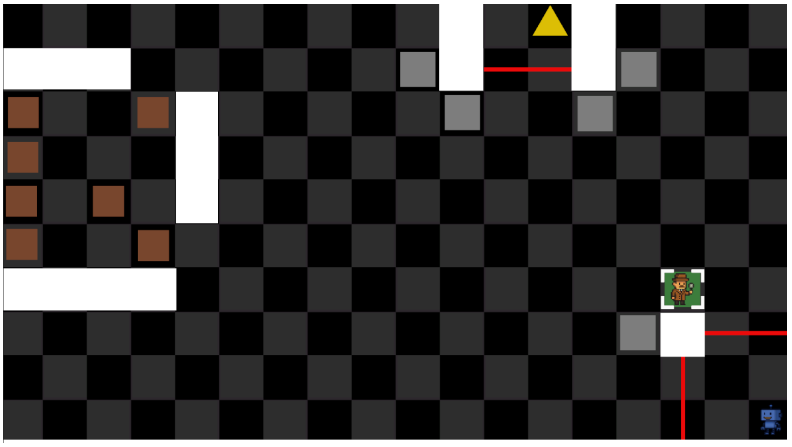
The game prototype consists of four levels. In the high-branching version:

- After Level 1, players can enter one of three states: talk, hack, or kill red bot.
- The outcome of Level 2 builds on the previous choice and again allows players to talk, hack, or kill red bot.
- In Level 3, players may choose to save, save and hack, or ignore/kill blue bot.
- Level 4 determines the final ending based on the accumulated decisions across all previous levels.

This structure results in 27 distinct narrative branches and 6 possible endings, including:

1. both robots die,
2. one robot survives without being hacked,
3. one robot survives but is hacked,
4. both survive and neither is hacked,
5. both survive and both are hacked, and
6. both survive with only one hacked.

In the low-branching version, hacked robots do not appear in subsequent levels, effectively making hack equivalent to kill. This reduces the overall complexity to 11 narrative branches and 3 final endings.



The third level from our research demo. The player is standing on one of two pressure plates to rescue a robot NPC.

BACKGROUND AND RELATED WORK

Scholars have long noted that when interactive narratives branch too extensively, they risk fragmenting story coherence and diminishing immersion, while purely linear structures restrict the player's sense of participation and agency (Ryan 2009). This tension highlights the challenge of maintaining both consistency and interactivity within narrative design, suggesting that an optimal level of branching may preserve engagement without losing narrative cohesion.

Even with developments in hypertext and interactive fiction, many games still deliver backstory and cutscenes through linear presentation, offering players limited control over narrative progression (Dickey 2005). Titles such as Façade (Procedural Arts 2005) and The Sims (Maxis 2005) demonstrate how increased player freedom can come at the cost of dramatic unity or emotional investment. These cases reinforce the need to balance choice and structure when designing branching narratives that remain meaningful.

Recent computational approaches have provided tools to analyze this balance more precisely. The Character Decision Points Detection (CHADPOD) framework, for example, models branching structures quantitatively and detects possible decision points in story graphs with high accuracy (Tikhonov 2024). Such methods make it feasible to measure branching complexity and to study how different levels of player-choice density might influence engagement and narrative flow.

Research on moral decision-making in games offers additional insight into how branching affects perception and emotion. Studies on ethical dilemmas and player response show how choices shape narrative consequence and emotional connection (Sicart 2013). Although moral choice is not the central variable in our research, these findings inform our understanding of how branching structures contribute to narrative consequence and player engagement, guiding the design considerations of our prototype.

A recent study examines systemic NPC-side branching in linear AAA games (Bostan 2022) through case studies of Disco Elysium (Studio ZA/UM 2019) and Hades (Supergiant 2020). Rather than relying only on explicit dialogue choices, these games use subtle NPC responses to player behavior, such as skill checks, item states, and repeated actions. This design enables a feeling of narrative reactivity and player recognition without the exponential content burden of large branching trees, thereby maintaining narrative coherence while supporting agency. Our prototype similarly includes extensive interaction with NPCs and explores lightweight systemic feedback within dialogue, where characters may acknowledge player actions or decisions across play sessions. By implementing a smaller-scale version of this approach, we investigate how subtle but persistent consequences can create meaningful agency and narrative impact in a constrained puzzle-narrative format.

The gameplay and visual direction of our research prototype draw inspiration from Baba Is You (Hempuli 2019), a minimalist 2D puzzle game that uses rule manipulation as its core mechanic. In Baba Is You, every word and object in the game world

functions as a movable block representing a piece of logic which allows the player to alter game rules directly within the level. This mechanic transforms linguistic statements into dynamic systems, creating a strong sense of discovery through experimentation. Our prototype adopts a similar 2D top-down grid layout and spatial logic structure, using simple tile-based environments to emphasize cause–effect clarity.

For puzzle design and pacing, Portal (Valve 2007) serves as another point of reference. Portal's level design introduces each mechanic through iterative experimentation, gradually layering complexity while maintaining strong spatial readability, guiding the player through problem-solving sequences that communicate both mechanical and narrative meaning.

The branching narrative framework in our prototype takes inspiration from Detroit: Become Human, which demonstrates how player choices can shape storylines through a visually explicit flowchart of decision paths. While our prototype operates on a smaller scale, it similarly integrates branching logic to explore how player agency interacts with narrative coherence.

METHODS

Hypothesis is that a balance point of the number of unique branches through a narrative is a moderate amount, in between the low branching and high branching versions of our demo.

Data Collection Instruments

For our quantitative methods, we will be using a subset of the PXI (Player Experience Index) for gauging immersion and autonomy for our players, and some custom questions for branching-related questions. We will be using these tools to assess the players' overall satisfaction of our game, and also how much they enjoyed the low/high branching versions of our game. All questions will be assessed with a Likert scale (1 - 7). See appendix for question examples. In addition to survey instruments, Participants will also take part in semi-structured interviews after gameplay.

We will also track which paths through the narrative tree the participants take by using a custom-designed branch key system that lets us uniquely identify all the different paths followed by each player. We will use this data to see which paths are taken most frequently. While this information isn't specifically relevant to answering our research question, it is still useful information in improving our game.

Research Protocol

The purpose of this study is to investigate how the number of narrative branches in Factory Reset influences players' satisfaction, experience, and enjoyment of our game.

For our sampling strategy, we will use convenience sampling, recruiting participants from friends, family, and schoolmates. We will be recruiting 23 participants. Before the study begins, participants will complete a consent form, also asking the participant how much they enjoy branching storytelling games. Participants that

strongly dislike or have never played branching narrative games (answering 2 or below on the screening question) will not participate in the study.

We will have 2 different experimental groups: Low Branching, and High Branching. To create these groups, we will balance participants between them based on their answer to the screening question in order to reduce potential biases.

We will be using a between-subjects design for our study. Each participant will only play the version they are assigned to. Low branching means there are very few options to solve each puzzle, which results in a lower number of branching paths, and vice-versa. The low branching version of the research demo will have 11 unique paths, while the high branching version will have 27.

We will have each participant play through the game for 15 minutes. In order to ensure that the participant experiences a reasonable amount of the available narrative, we will have the participant play longer than 15 minutes if they have not completed a minimum number of unique paths. For the high-branching treatment group, they must experience at least 11 unique paths, playing for longer than 15 minutes if they have not. For the low-branching treatment group, they must experience at least 4 unique paths.

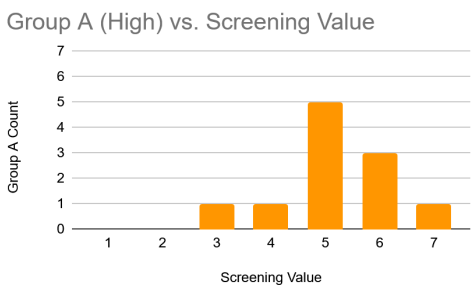
After collecting all data, we will be conducting an unpaired t-test to compare the mean of each individual construct in our survey.

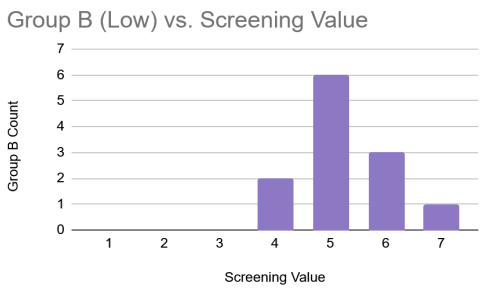
We will also conduct semi-structured interviews after the participants have filled out the survey, asking open ended questions regarding their feelings of agency throughout the game and their thoughts on the narrative design. We will use the qualitative nature of the data gathered to provide further context and insight into our quantitative findings, as well as hear the reasoning behind their responses. This mixed-method approach enables a better understanding of our findings and potentially reveals nuance that would otherwise be missed through purely quantitative measures

RESULTS

Participants

In total, we recruited 23 participants. We had 11 in Group A, our high branching group, and 12 participants in Group B, our low branching group. We distributed the participants based on their response to the screening question regarding their level of enjoyment of branching narrative games as shown below.





Survey Data

We had 5 main constructs, measuring the participants' experience via the post-game survey. These constructs include:

1. Overall Enjoyment (general questions about the player's enjoyment of the game)
2. Perceived Branching Complexity (how complex the player thought the branching system was in the game)
3. Autonomy (how much agency and control the player felt when playing the game)
4. Too much Branching (asking the players if they felt overwhelmed/thought there were too many branches)
5. Immersion (how immersed the players felt while playing the game)

After conducting an unpaired two-tailed t-test for each construct (averaging the responses to all questions in each construct), we found that only one construct had statistically significant data: Perceived Branching Complexity. We calculated the average of each participant's responses for these questions:

1. My choices led to noticeably different story paths.
2. The number of branches in the story felt high.
3. The branches I encountered felt meaningful.

After conducting the t-test based on these results, we found these results:

	Group 1 Mean	Group 2 Mean	p-value	Group 1 standard deviation	Group 2 standard deviation
Overall Enjoyment	5.55	5.54	0.99	0.688	0.753
Perceived Branching Complexity	5.27	4.19	0.0049	0.727	0.893
Autonomy	6.03	5.36	0.0669	0.795	0.859
Too Much Branching	2.51	1.72	0.127	1.53	0.777
Immersion	4.88	4.75	0.705	0.687	0.89

T-Test Results - Figure 1

Group 1 was the high branching group while Group 2 was the low branching group. With a p-value of 0.0049, we have statistically significant evidence to show that there is a significant difference between the two groups' averages. We see that Group 1's average is around 5.2, while Group 2's average is around 4.2 (Group 1's participants answered the above 3 questions on average 1 point higher than Group 2's participants did).

Similarly, we also compared the difference of results between these 2 questions individually:

1. How many different story branches did you notice? (number input)
2. How many endings do you think are in our game? (number input)

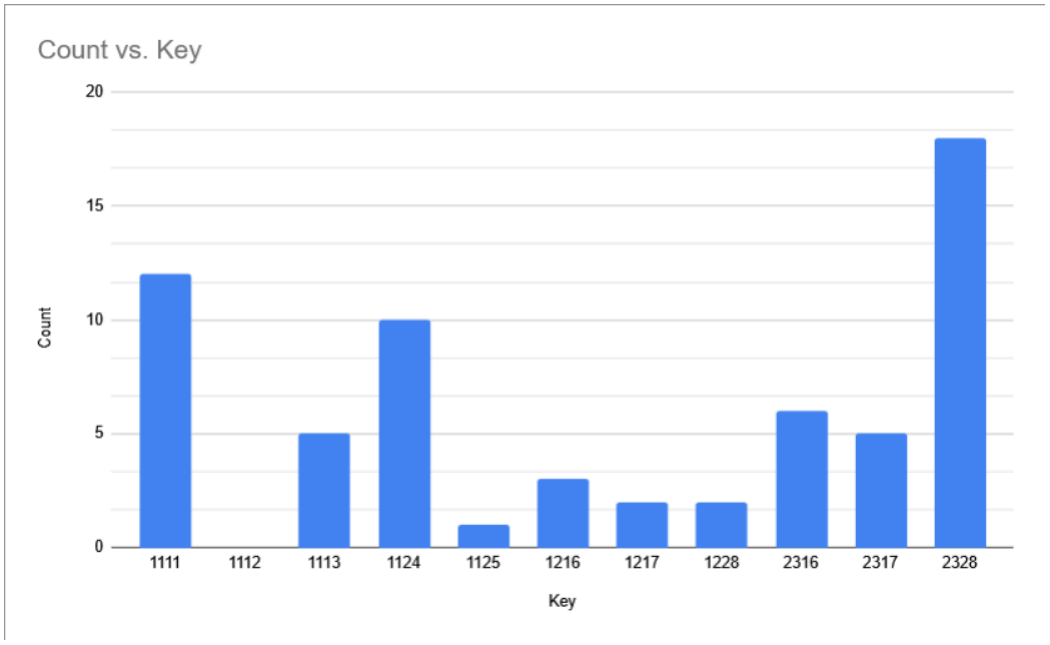
We found that the results to both survey questions had statistically significant data as well. The first question had a difference of about 2.68 points (Group 1 mean: 6.45, Group 2 mean: 3.77) with a p-value of 0.0075. The second question had a difference of about 1.74 (Group 1 mean: 5.45, Group 2 mean: 3.71) with a p-value of 0.017. This means that we have evidence to prove that the mean of the answers between the 2 groups are statistically significant for both of the above questions (Group 1 thought there were more story branches and endings than Group 2 did).

All of our other constructs did not show statistically significant data for the difference in answers between the 2 groups.

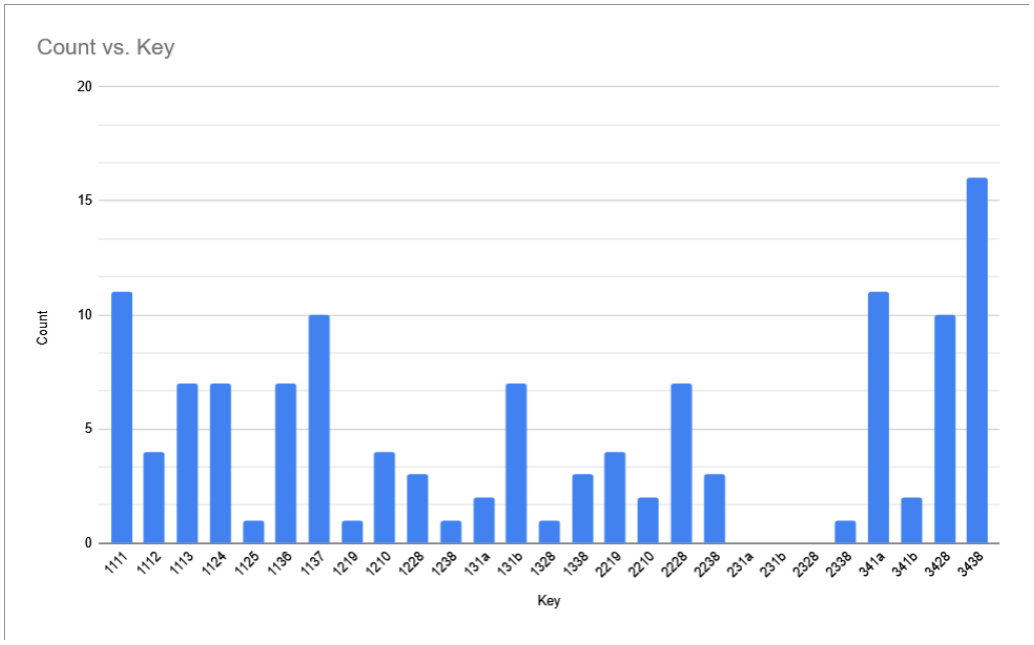
Branch Data

We also collected information on the specific paths through which our participants played through the demo. We collected this information using our branch key system, where each unique path through the given narrative tree has its own 4 digit key to identify it, as described in our Research Protocol.

For all of our participants, through both treatment groups, we had 189 individual runs through the entire game. The frequencies that each key was played is shown below, split by treatment group, as the individual groups had separate branch keys.

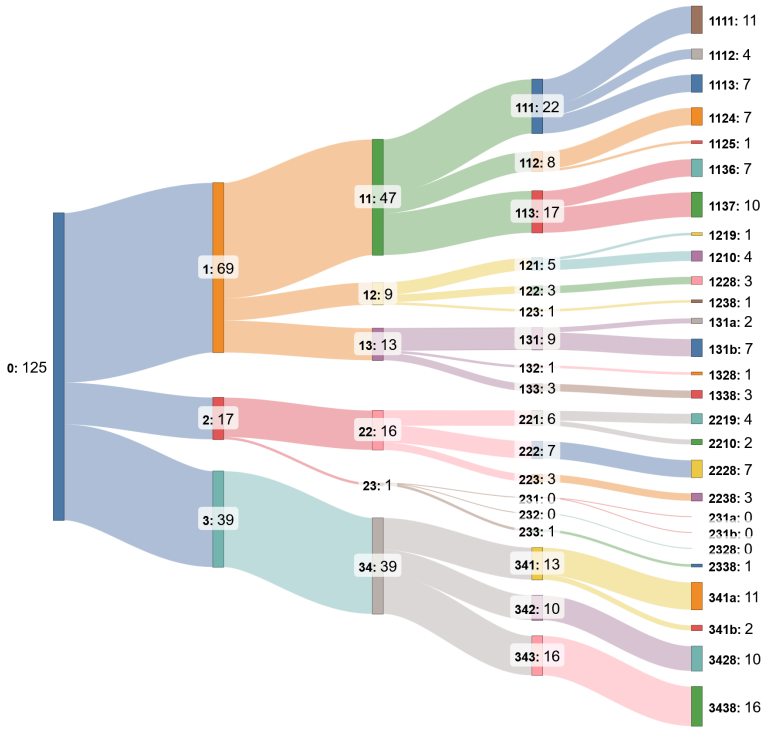


Treatment Group B Key Frequency Graph

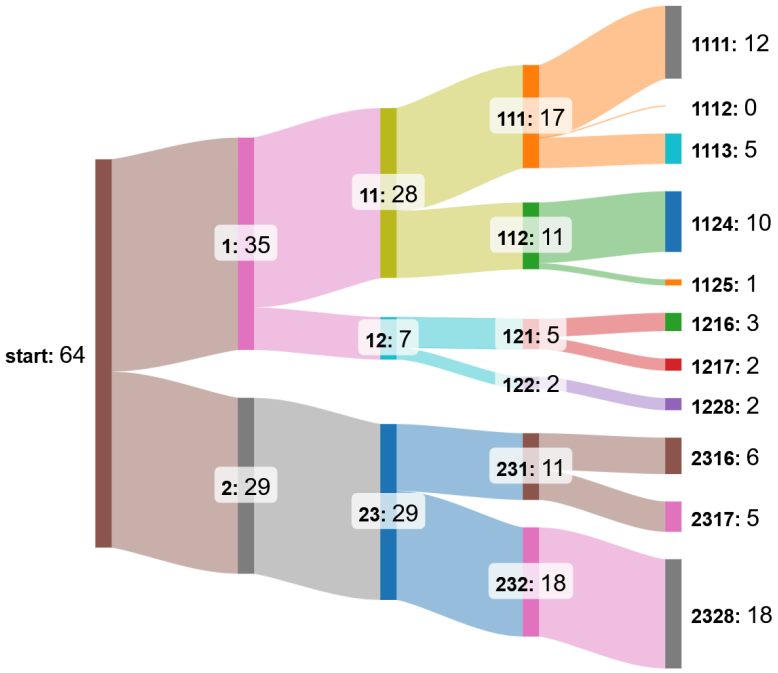


Treatment Group A Key Frequency Graph

A different visualization of this data is shown below through two Sankey flow diagrams. These are again split by treatment group, and display the branching nature of the game. They also demonstrate at each individual decision point where a majority of participants decided to play and where fewer participants chose. Each diagram follows the flow of decisions in gameplay, and subsequently builds the branch key for that path at each node.



Treatment Group A, High Branching Sankey Flow Diagram. (**Key:** # of participants)



Treatment Group B, Low Branching Sankey Flow Diagram (**Key:** # of participants)

DISCUSSION

Survey Data

From our data, we conclude that the only statistically significant construct was the Perceived Branching Complexity construct. This is to be expected, as one of the independent variables we directly controlled for was the number of paths available in each version of our game. Although the rest of our constructs did not contain statistically significant data, it is interesting to note that the Overall Enjoyment and the Immersion constructs had the highest p-values, denoting that the different versions of our game affected overall enjoyment and immersion of our game the least. The Too much Branching construct's results are also expected, as the higher branching group is expected to have more participants report that there is too much branching in their version of the game than in the low branching group.

Finally, the last observation we made was that the Autonomy construct had a p-value of 0.0669. Although not statistically significant, we can extrapolate that the participants in the higher branching group may have felt a higher sense of autonomy due to the larger number of choices to select from in each level than the participants in the lower branching group.

Based on these findings, we came to the conclusion that for our game Factory Reset, an optimal level of branching in the narrative is around the amount of branching in the high-branching variation of our research demo, and not in between the two versions. While there was roughly a 15% increase in the branching satisfaction survey responses in the high branching version, this came at the cost of roughly a doubling of complexity (11 unique paths in the low branching version compared to 27 unique paths in the high branching version).

However, when accounting for a number of statements collected in our semi-structured interviews, we believe that this extra development cost is worth it. A number of participants that played the lower branching version expressed a feeling of wanting more. Participant G stated how the branches that did exist felt good but felt more were needed to feel coherent. To create the low branching version, we condensed the high branching version by combining the branches for hacking and destroying a robot, which Participant E directly noted felt off in their interview.

Branch Data

In the branch key data, the most frequent paths taken were 3438 in high branching and 2328 in low branching, which are both the 'lonely' route where the player ends the game after destroying or not rescuing all robots. In low branching, the next most frequent key is 1111, and in high branching tied for next most frequent is also 1111, which for both groups are the routes where all robots are alive and are friendly with the player. This suggests that players are more likely to first explore one of the extremes, since these routes are the most extreme ones.

Looking at the next most frequent, in high branching, keys 341a and 1137 which are both routes that end with the same ending: one robot was destroyed and the other remains friendly with the player. The difference between these two is which of the two robots the player destroyed and kept around. Similarly in low branching, the

next most frequent is 1124, which results in the same ending as the routes in the high branching keys: one robot alive and friendly and the other destroyed. This route can be described as a 'middling' route since it is balanced on the spectrum of the robots being either alive or destroyed. However, of the routes that could be described this way, it still is the most extreme of these 'middling' routes as there are routes where robots are friendly or unfriendly towards the player and not the extreme of being alive or destroyed, thus furthering our belief that players tend toward more extreme choices.

A possible explanation for this tendency is that it is part of a strategy the players employ in order to attempt to explore more of the narrative tree. A large number of our participants noted during gameplay that on various runs they were actively trying for a specific ending or route. For example, Participant E stated on their third run after just reaching an ending with both robots ending friendly towards the player that "now it's time for the bad ending" and proceeded to make choices that they felt led in that direction. We noticed that players very rarely played paths that involved them making contradicting choices (being amicable towards the robots and then being hostile, and repeating this flip flopping). Because of this, we feel this furthers the explanation of an exploration strategy, as playing through a route with 'flip-flopping' decision making wouldn't feel like it might lead as strongly to a desired ending.

Future Development

Besides the feedback we received for the number of available branches built into our game, we received many open-ended comments and requests for the next iteration of our game. Some comments include: better controls for the player (being able to hold down arrow keys), having the NPC's talk to the player, then having the player choose to cooperate or hack the robot, changing the puzzle order of level 4 for every run, and many, many more ideas. We will do our best to implement a lot of this feedback into our game next semester.

Limitations and Future Work

While the study produced valuable insights, there are still aspects that could be further improved or expanded in future research. First, due to time and scope constraints, we did not incorporate biometric measures of engagement, such as heart rate variability, galvanic skin response, or eye-tracking data. While such measures can offer continuous and objective indicators of emotional arousal, applying them meaningfully would require a more mature and fully developed game prototype than the one used in this study. As a result, we relied on PXI self-report measures and qualitative interviews, which, although valuable, depend on participants' retrospective reflection. Future work could explore biometric monitoring in an in-person laboratory setting once the game has progressed to a more advanced stage, allowing for richer triangulation of player experience.

Second, the research protocol required participants in the high-branching treatment group to complete a minimum of eleven unique narrative branches. Our internal test runs indicated that this target could be reached in under ten minutes, so we extended the allotted playtime to fifteen minutes to accommodate participants who would be experiencing the game for the first time. However, our observations during

the study revealed that first-time players often required significantly longer than anticipated to navigate and discover eleven distinct branches. This mismatch between task complexity and allocated playtime may have introduced variability in completion rates and decision-making patterns. Future work should consider adjusting the target number of required branches, extending the play duration, or designing an adaptive branching structure that ensures all participants can experience the intended branches within a reasonable timeframe.

Finally, our study found that the concepts of enjoyment and immersion in our game were not statistically significant between the 2 groups of players. This is not too surprising since our independent variable, branching complexity, is not the only factor that can affect the player's enjoyment and immersion. In order to improve the overall enjoyment and immersion for our players, we can add more meaningful NPC interactions, better assets, and smoother controls.

CONCLUSION

In examining how the density of narrative pathways shapes player engagement, this study demonstrates that the structure of branching stories meaningfully affects the player experience. By comparing high-branching and low-branching versions of our prototype under controlled playtime, we observed clear differences in how players perceived agency, coherence, and emotional involvement. The PXI results, supported by exit interview responses, suggest that the participants in the high-branching group felt more satisfied with the number of branches in our game than the participants in the low-branching group. This research therefore provides actionable insights for designers seeking to create meaningful, choice-driven interactive narratives.

REFERENCES

Bostan, Barbaros, ed. Games and Narrative : Theory and Practice. Cham: Springer, 2022. https://doi.org/10.1007/978-3-030-81538-7.

Dickey, Michele D. 2005. "Engaging by Design: How Engagement Strategies in Popular Computer and Video Games Can Inform Instructional Design." Educational Technology Research and Development 53 (2): 67–83. https://www.jstor.org/stable/30220429.

Ryan, Marie-Laure. 2009. "From Narrative Games to Playable Stories: Toward a Poetics of Interactive Narrative." StoryWorlds: A Journal of Narrative Studies 1 (1): 43–59. https://www.jstor.org/stable/25663007.

Sicart, Miguel. 2013. "Moral Dilemmas in Computer Games." Design Issues 29 (3): 28–37. https://www.jstor.org/stable/24267087.

Tikhonov, Alexey. 2024. Branching Narratives: Character Decision Points Detection. https://doi.org/10.48550/arxiv.2405.07282.

Game References

Hempuli. 2019. Baba Is You. PC Game. Hempuli.

Larian Studios. 2023. Baldur's Gate 3. PC Game. Larian Studios.

Maxis. 2000. The Sims. PC Game. Maxis.

Procedural Arts. 2005. Façade. Interactive Drama Game. Procedural Arts.

Quantic Dream. 2018. Detroit: Become Human. Console Game. Quantic Dream.

Studio ZA/UM. 2019. Disco Elysium. PC Game. Studio ZA/UM

Supergiant Games. 2020. Hades. PC Game. Supergiant Games.

Toby Fox. 2015. Undertale. PC Game. Toby Fox.

Valve Corporation. 2007. Portal. PC Game. Valve Corporation.